Steven Hoekstra

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Profile

I study how heterogeneous households and firms, boundedly rational expectations and market interactions shape monetary and fiscal policy transmission. I connect heterogeneous-agent equilibrium models and agent-based models, using neural-network solvers, simulation-based inference and microdata to assess economic mechanisms and the reliability of policy predictions.

Interests

Macroeconomics, heterogeneous-agent models, agent-based modelling, expectations and bounded rationality, policy evaluation, networks, simulation-based inference, high-performance computing.

Education

PhD in Economics — University of Amsterdam 2023 — current
Supervisors: Prof. dr. C.H. Hommes and Prof. dr. F.P. Pijpers
Expected thesis submission: May 2027; expected defence: September 2027

MPhil Economics and Econometrics — Tinbergen Institute *(cum laude)* 2021 — 2023
Specialisation: Complexity, Mathematics, and Computation
Thesis: An Agent Based Model of the Dutch Economy
Supervisors: Prof. dr. C.H. Hommes and Prof. dr. F.P. Pijpers

MSc Econometrics — University of Amsterdam *(cum laude)* 2020 — 2021
Specialisation: Complexity and Economic Behaviour
Thesis: Endogenous Financial Instability via Heterogeneous Beliefs
Supervisor: Prof. dr. C.H. Hommes

MSc Economics — London School of Economics 2019 — 2020
Specialisation: Public Economics
Thesis: The Inequality of Education using a Two Agent New Keynesian Model
Supervisor: Prof. dr. F. Cowell

Exchange Programme — Royal Melbourne Institute of Technology 2018 — 2019
Minor in Finance and Programming

BSc Economics and Business Economics — University of Groningen 2016 — 2019
Thesis: The Impact of News Coverage on Security Prices

BSc Econometrics — University of Groningen 2015 — 2018
Thesis: Arbitrage Smoothing and Term Structure Models

Working Papers and Research Software

Forecasting with a Dutch Agent-Based Model and Optimally Behaving Households
Working paper, with Cars Hommes and Frank Pijpers. PDF
Runner-up, Society for Computational Economics Graduate Student Paper Award, CEF 2026

BeforeIT.jl: High-Performance Agent-Based Macroeconomics Made Easy
Research software; package co-author. Developed the validation suite.
Package · Original preprint (2025)

Work in Progress

When Do Estimated Macroeconomic Models Misread Wage–Price Transmission?
Single-author. Known-truth experiments on misspecification and intervention-response reliability; Dutch wage–price transmission and specification sensitivity during the 2022 inflation episode.

Shortage or Misallocation? The Welfare Loss from Dutch Housing Policy, 2015–2035
Integrating housing, labour mobility, demography and firm dynamics in a macroeconomic ABM; calibration and welfare comparisons are ongoing.

Presentations

2026	Computing in Economics and Finance (CEF), Society for Computational Economics
2026	Workshop on Economic Science with Heterogeneous Interacting Agents (WEHIA)
2025	Agent-Based Modelling for Policy (ABM4Policy), European Central Bank, Frankfurt
2025	7th Behavioral Macroeconomics Workshop, University of Bamberg
2025	MacroABM 2nd Workshop, *Microdata in Macromodels*, Vienna University of Economics and Business
2025	Workshop on Economic Science with Heterogeneous Interacting Agents (WEHIA)
2025	De Nederlandsche Bank, Amsterdam
2023	De Nederlandsche Bank, Amsterdam

Work Experience

De Nederlandsche Bank — PhD Intern Apr. — Dec. 2025
· Implemented the Bank of Canada CANVAS agent-based model for the Netherlands
· Estimated inflation channels using neural ratio estimation
· Wrote a note on the wage-price channel using this estimated model

Statistics Netherlands (CBS) — Thesis Intern Mar. — Sept. 2023
· Calibrated the agent-based model of Poledna et al. (2023) with Dutch register and national accounts data
· Developed an extension incorporating optimal consumption
· Ran simulations with both models and analysed their internal dynamics

SEO Amsterdam Economics — Research Assistant 2020 — 2021
· Researched academic pressure among secondary school students
· Conducted stakeholder interviews and set up a nationwide survey
· Performed statistical analysis on the survey outcomes

Research Computing Allocations

Snellius supercomputer
Three awarded allocations of one million CPU hours each (three million CPU hours in total).

Teaching

Graduate

Behavioural Macro and Finance — Lecturer 2025
Developed and taught the MSc lecture series with Cars Hommes, University of Amsterdam.

MPhil guest lectures — Tinbergen Institute
Annual guest lectures in Behavioural Macroeconomics and Complexity.

Undergraduate

BSc Econometrics thesis supervision — University of Amsterdam
Ongoing.

Mathematics 3: Advanced Linear Algebra — Tutor 2025 — 2026

Mathematics 2: Linear Algebra — Tutor 2024 — 2025

Microeconomics for AE — Tutor 2023 — 2024

Achievements

Honours

Cum Laude — MPhil Economics and Econometrics, Tinbergen Institute 2023

Cum Laude — MSc Econometrics, University of Amsterdam 2021

Scholarships

Full scholarship (merit based) — Tinbergen Institute 2022

Tuition fee waiver — Tinbergen Institute 2021 — 2023

Visits and Summer Schools

Visiting Scholar — IIASA, Laxenburg Aug. 2023
International Institute for Applied Systems Analysis. Visited Dr. S. Poledna to work on agent-based models

Tools for Macroeconomists: Advanced Tools — University of Oxford Aug. 2021
Doctoral-level course on DSGE models: solving and simulating models with heterogeneous agents, continuous time models, occasionally binding constraints and bounded rationality

Tools for Macroeconomists: The Essentials — University of Oxford Aug. 2021
Doctoral-level course on DSGE models: numerical analysis, perturbation and projection methods, Blanchard-Kahn conditions and sunspot solutions, the Kalman filter and Bayesian estimation

Skills

Programming	Matlab, Python, R, Julia, C, SQL, Bash
High-performance computing	Parallel computing (MPI, BSPlib)
Methods	Agent-based modelling, heterogeneous-agent models, neural-network solvers, simulation-based inference, network methods, machine learning, econometrics, dynamic stochastic general equilibrium modelling
Packages	Dynare, Stata

References

Prof. dr. C.H. Hommes
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Prof. dr. F.P. Pijpers
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